

Problem Set 6

Ryan T. Moore

1 Multiarm Bandits

- a) Demonstrate how treatment arms' posterior probabilities of being the best become more and less certain as the number of trials change in absolute terms and in relative terms. See the code provided in `06-bandits.R` for help.
- b) Replicate the three left panels of Figure 1 from Offer-Westort, Coppock, and Green (2021). See the code provided in `06-bandits.R` for help.
- c) Suppose that financial and IRB constraints are not binding. Describe a political or policy experiment that you would like to run via a multiarm bandit.

References

Offer-Westort, Molly, Alexander Coppock, and Donald P. Green. 2021. "Adaptive Experimental Design: Prospects and Applications in Political Science." *American Journal of Political Science* 65 (4): 826–44. https://alexandercoppock.com/offer-westort_coppock_green_2021.pdf.